

## CENTRAL ILLINOIS, IL, USA

WMO: 724397

Lat: 40.479N

Lon: 88.915W

Elev: 865

StdP: 14.24

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 54831

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -1.0       | 3.8 | -8.6                          | 3.6 | 0.9  | -4.2 | 4.5 | 4.9  | 32.3                  | 28.6 | 28.8 | 31.7 | 10.7                  | 310  | 0.583 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 18.6                   | 91.3            | 75.7 | 89.4 | 74.9 | 86.9 | 73.4 | 79.1                | 87.6 | 77.4 | 85.3 | 75.8 | 83.5 | 12.4                 | 210  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 76.8                            | 144.3 | 84.5 | 75.0 | 135.7 | 82.6 | 73.3 | 128.0 | 80.3 | 43.2          | 87.5 | 41.4 | 85.7 | 39.8 | 83.6 | 85.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |       |            |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|-------|------------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |       | n=50 years |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max   | Min        | Max   |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)   | (o)        | (p)   |
| 27.8              | 24.9 | 22.3 | -8.4                       | 95.2 | 6.9                | 3.2 | -13.4                                              | 97.5 | -17.4      | 99.4 | -21.3      | 101.2 | -26.3      | 103.5 |
|                   |      |      | -9.3                       | 82.1 | 6.8                | 1.8 | -14.2                                              | 83.4 | -18.2      | 84.5 | -22.0      | 85.4  | -27.0      | 86.7  |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |       |            |       |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |       |            |       |

## Monthly Climatic Design Conditions

|                                                                   |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                   |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 52.0   | 24.9  | 27.9  | 40.3  | 52.3  | 63.0  | 72.2  | 75.0  | 73.2  | 67.3  | 54.4  | 41.3  | 30.4  |
|                                                                   | DBStd       | 20.01  | 12.10 | 11.52 | 11.54 | 9.76  | 8.96  | 6.05  | 5.54  | 5.19  | 7.60  | 9.72  | 9.91  | 10.44 |
|                                                                   | HDD50       | 2807   | 780   | 622   | 344   | 86    | 8     | 0     | 0     | 0     | 1     | 63    | 292   | 611   |
|                                                                   | HDD65       | 5813   | 1244  | 1039  | 768   | 395   | 150   | 12    | 3     | 4     | 62    | 353   | 710   | 1073  |
|                                                                   | CDD50       | 3525   | 2     | 3     | 44    | 154   | 410   | 665   | 774   | 718   | 520   | 200   | 32    | 3     |
|                                                                   | CDD65       | 1053   | 0     | 0     | 2     | 12    | 87    | 227   | 311   | 258   | 131   | 25    | 0     | 0     |
|                                                                   | CDH74       | 9457   | 0     | 0     | 19    | 140   | 798   | 2054  | 2951  | 2132  | 1156  | 206   | 1     | 0     |
|                                                                   | CDH80       | 3208   | 0     | 0     | 1     | 23    | 231   | 711   | 1104  | 698   | 396   | 44    | 0     | 0     |
| Wind                                                              | WSAvg       | 11.2   | 13.0  | 12.5  | 12.9  | 13.3  | 11.5  | 10.0  | 8.5   | 7.8   | 8.7   | 11.1  | 12.5  | 12.3  |
| Precipitation                                                     | PrecAvg     | 37.8   | 2.4   | 1.9   | 2.6   | 3.6   | 4.6   | 4.2   | 3.9   | 3.7   | 2.4   | 3.2   | 3.1   | 2.1   |
|                                                                   | PrecMax     | 52.6   | 6.5   | 5.4   | 5.4   | 7.2   | 9.8   | 8.6   | 12.1  | 6.2   | 7.6   | 6.2   | 6.5   | 6.0   |
|                                                                   | PrecMin     | 32.7   | 0.7   | 0.4   | 1.0   | 1.5   | 0.7   | 1.1   | 1.6   | 1.4   | 0.7   | 1.2   | 0.4   | 1.1   |
|                                                                   | PrecStd     | 5.5    | 1.3   | 1.3   | 1.2   | 1.4   | 2.4   | 2.0   | 2.5   | 1.4   | 1.7   | 1.6   | 1.5   | 1.2   |
| Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 59.4  | 63.1  | 77.0  | 83.0  | 89.5  | 92.9  | 97.1  | 93.6  | 92.8  | 84.9  | 71.1  | 61.0  |
|                                                                   |             | MCWB   | 56.8  | 56.5  | 63.7  | 64.9  | 71.5  | 73.6  | 76.2  | 77.1  | 73.2  | 70.4  | 59.2  | 58.4  |
|                                                                   | 2%          | DB     | 52.4  | 55.5  | 68.8  | 78.5  | 85.7  | 90.2  | 91.3  | 90.1  | 88.3  | 79.5  | 64.9  | 54.8  |
|                                                                   |             | MCWB   | 49.7  | 50.7  | 59.8  | 62.6  | 68.6  | 73.6  | 77.8  | 76.2  | 72.6  | 66.0  | 58.0  | 51.9  |
|                                                                   | 5%          | DB     | 46.4  | 48.8  | 63.5  | 73.3  | 82.3  | 87.6  | 89.0  | 86.9  | 84.7  | 75.1  | 61.0  | 49.6  |
|                                                                   |             | MCWB   | 43.8  | 44.5  | 56.2  | 60.3  | 67.6  | 72.8  | 76.4  | 74.4  | 70.1  | 64.2  | 55.6  | 46.7  |
|                                                                   | 10%         | DB     | 40.8  | 43.7  | 58.4  | 69.2  | 78.6  | 84.1  | 86.1  | 83.8  | 81.2  | 70.2  | 56.7  | 45.1  |
|                                                                   |             | MCWB   | 38.3  | 39.8  | 52.3  | 57.9  | 65.9  | 71.3  | 74.5  | 72.7  | 67.8  | 61.7  | 51.6  | 42.3  |
| Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 57.8  | 57.9  | 64.9  | 68.2  | 74.6  | 78.9  | 82.2  | 80.5  | 77.0  | 72.9  | 63.3  | 58.5  |
|                                                                   |             | MCDB   | 59.5  | 60.5  | 73.5  | 78.4  | 84.6  | 87.6  | 90.5  | 88.9  | 87.2  | 82.0  | 66.6  | 59.9  |
|                                                                   | 2%          | WB     | 50.3  | 51.3  | 61.3  | 65.0  | 72.2  | 77.0  | 79.9  | 78.4  | 74.9  | 69.7  | 59.7  | 52.5  |
|                                                                   |             | MCDB   | 51.9  | 55.0  | 66.7  | 73.6  | 81.4  | 85.8  | 88.2  | 86.8  | 84.3  | 76.2  | 63.3  | 53.8  |
|                                                                   | 5%          | WB     | 43.8  | 45.4  | 58.0  | 62.2  | 70.1  | 75.2  | 78.1  | 76.6  | 72.9  | 66.6  | 56.4  | 46.8  |
|                                                                   |             | MCDB   | 46.2  | 48.4  | 63.1  | 70.6  | 78.3  | 83.1  | 85.7  | 84.0  | 80.8  | 71.2  | 60.3  | 48.7  |
|                                                                   | 10%         | WB     | 38.2  | 40.1  | 53.0  | 59.7  | 68.1  | 73.3  | 76.3  | 74.8  | 70.8  | 63.4  | 52.2  | 42.1  |
|                                                                   |             | MCDB   | 40.4  | 43.6  | 57.6  | 67.8  | 75.6  | 81.6  | 83.7  | 81.5  | 77.8  | 69.3  | 56.4  | 44.5  |
| Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 14.7  | 16.0  | 18.3  | 20.8  | 20.3  | 19.2  | 18.6  | 18.8  | 21.1  | 19.0  | 16.8  | 13.6  |
|                                                                   |             | MCDBR  | 20.0  | 22.6  | 24.3  | 26.4  | 23.8  | 22.2  | 20.2  | 21.4  | 24.0  | 24.4  | 22.1  | 18.8  |
|                                                                   |             | MCWBR  | 17.4  | 17.6  | 15.8  | 14.7  | 11.3  | 10.1  | 9.5   | 10.1  | 10.0  | 12.7  | 16.0  | 17.0  |
|                                                                   | 5% WB       | MCDBR  | 19.4  | 21.6  | 22.1  | 23.0  | 21.0  | 18.4  | 18.2  | 18.4  | 20.2  | 19.7  | 20.3  | 18.0  |
|                                                                   |             | MCWBR  | 17.9  | 18.1  | 15.8  | 14.6  | 11.6  | 9.8   | 10.2  | 10.2  | 9.7   | 12.5  | 16.4  | 17.1  |
| Clear-Sky Solar Irradiance                                        | taub        |        | 0.294 | 0.307 | 0.344 | 0.387 | 0.426 | 0.438 | 0.452 | 0.443 | 0.403 | 0.353 | 0.309 | 0.292 |
|                                                                   | taud        |        | 2.437 | 2.398 | 2.383 | 2.269 | 2.213 | 2.224 | 2.169 | 2.182 | 2.270 | 2.415 | 2.493 | 2.459 |
|                                                                   | Ebn at Noon |        | 276   | 288   | 288   | 281   | 271   | 267   | 262   | 261   | 264   | 268   | 269   | 268   |
|                                                                   | Edh at Noon |        | 25    | 30    | 34    | 41    | 45    | 44    | 46    | 45    | 38    | 30    | 24    | 23    |
| All-Sky Solar Radiation                                           | RadAvg      |        | 573   | 858   | 1164  | 1507  | 1730  | 1972  | 1972  | 1733  | 1471  | 993   | 661   | 481   |
|                                                                   | RadStd      |        | 66    | 80    | 87    | 120   | 140   | 135   | 101   | 109   | 113   | 113   | 57    | 62    |

## Historical Trends

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |       | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (45 neighbors) |       | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +151  | N/A   |

Nomenclature: See separate page